# Guarded summation-polynomial fall degrees

Study report, 28 October 2026. Derived bounds and development checks; no performance claim.

Let $S$ be $B$ distinct rational affine-point abscissas, $g(T) = \prod_{s \in S}(T - s)$, and $r = x(R)$ for an affine rational target. The ordinary polynomial-ring inputs are

$$g(x), \quad g(y), \quad g(z), \quad S_3(x, y, u), \quad S_3(u, z, r), \quad v(x - y) - 1.$$

Both point signs are allowed. The guard requires distinct first abscissas. Every degree below is ordinary total degree with explicit membership equations. The first-fall convention is homogeneous-top syzygies modulo Koszul syzygies.

| Setting, unsatisfiable instances | First fall | Last fall |
|---|---|---|
| Short Weierstrass, characteristic $> 3$, $B = 3$ | 4 | 5 |
| Same model, $B \geq 3$ | $B + 1$ | $\max(5, B + 1) \leq d_F \leq 3B - 4$ |
| Ordinary binary, $B = 3$, $g(T) = T^3 + c$ | 4 | 4 |
| Ordinary binary, $B = 3$, other split cubics | 4 | 5 |
| Ordinary binary, $B \geq 4$ | $B + 1$ | $B + 1 \leq d_F \leq 3B - 4$ |

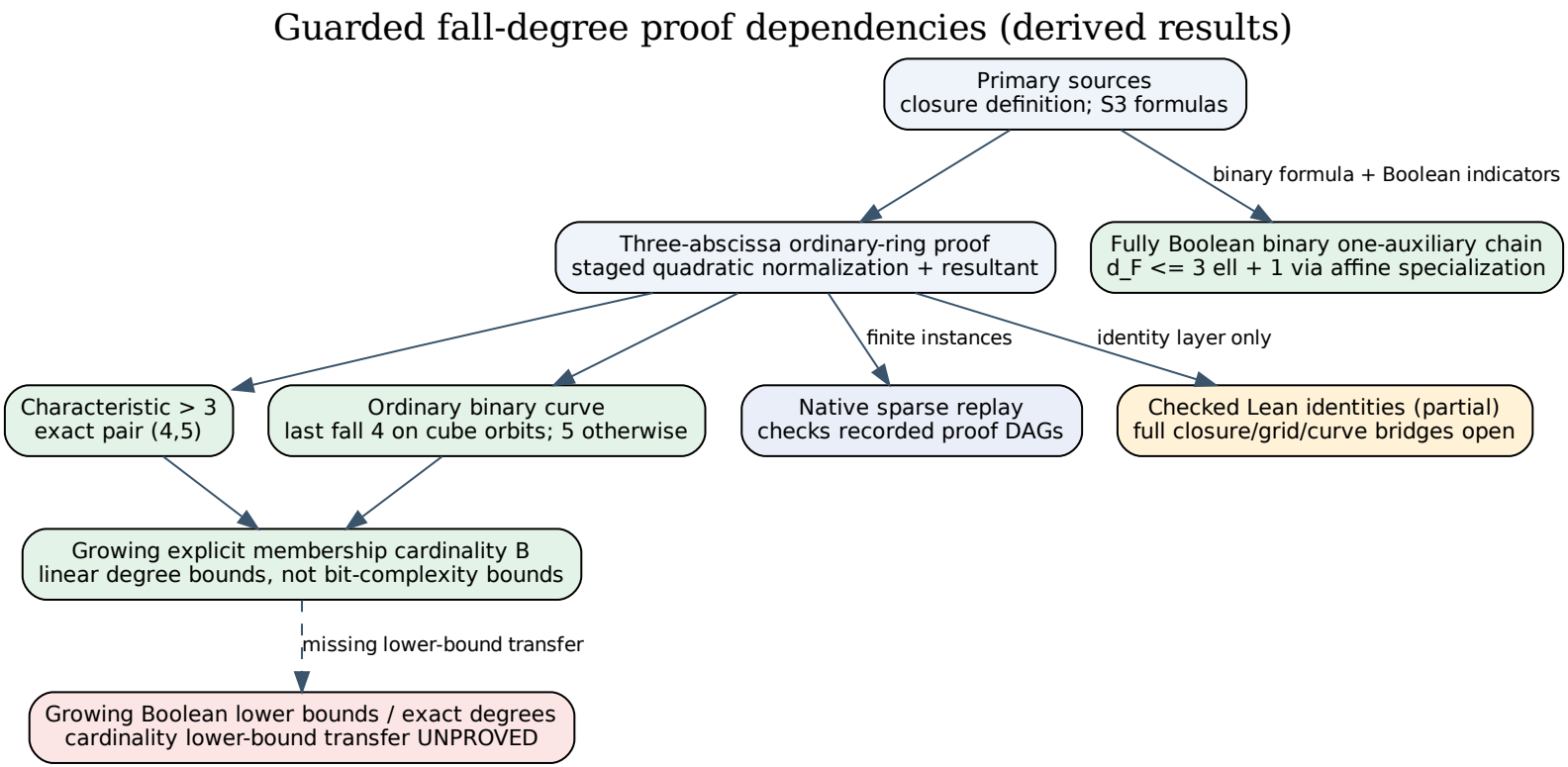
The binary curve is $Y^2 + XY = X^3 + a_2X^2 + a_6$, $a_6 \neq 0$. Its cube-orbit exception requires even extension degree, so the pair is $(4, 5)$ for all three-abscissa unsatisfiable instances over $\mathbb{F}_{2^n}$ with odd $n$. The growing lower bound depends on the degree-$B$ membership input and gives no lower bound in the variable count of a fixed-degree Boolean encoding.

**Direct Boolean upper bound.** For the binary two-$S_3$ chain with one shared $u$, three linear $\ell$-bit base embeddings, and every Boolean bit equation explicit, $d_F \leq 3\ell + 1$ for $\ell \geq 1$, independently of $n$. This holds with or without the inverse guard, without rational-lift assumptions. Fixing the base bits leaves an affine auxiliary system; staged assignment indicators localize its contradiction. Proof size may be exponential.

**Method.** Staged multiplication and cubic reduction normalize two quadratic fibres, then derive their scalar resultant within degree five. Finite-grid inversion yields 1. Characteristic two uses a Frobenius identity instead of division by two. Closed separating superspaces give the lower bounds. An exact triple-selector partition lifts local proofs to growing cardinality.

**Verification boundary.** Native sparse replay checks the compact historical proof DAGs against separately supplied inputs. Native binary-field arithmetic and the pinned cube-orbit fixture have development unit checks; no new field/base/target sweep or timing measurement was run. Lean checks only the declarations and hypotheses listed in `formal/PROOF_MAP.json`. The complete filtered-closure, grid-inversion, Boolean-localization, and elliptic-curve bridges are not formalized.

**Prior work.** The checked sources are Huang–Kosters–Yang–Yeo, 1505.02532; Caminata–Gorla, 2112.05579; Semaev, 1504.01175; and Kosters–Yeo, 1503.08001. Assumptions and repository links are in `SOURCES.md`. No literature-priority claim is made.




Solid arrows denote dependencies in the written arguments. The dashed arrow is an unresolved lower-bound transfer, not a proved implication. The Lean box certifies algebraic identity lemmas only; native replay certifies finite proof operations only. Neither substitutes for the complete mathematical derivation.

The existing E12 Boolean Weil-descent records are unchanged. Its completed cell retains labels $\geq 5, 4, \geq 5, 4$, a frozen lower median of 4, and a separate independent closure-replay gap. The present coefficient-field theorems do not close that gap. The direct Boolean upper theorem requires a separate encoding audit before application to that ledger. No performance or security claim follows.

**Reproducible records.** See `README.md`, the proof files, `native/validation/`, `identities/`, and `formal/`. The editable diagram source is `reports/proof-dependencies.dot`. Historical Python sources are preserved as provenance, not as an execution path.